as follows is a metric on the product $X_1 \times X_2 \times \cdots \times X_n$: $d(\{x_i\},\{y_i\}) = \max d_i(x_i,y_i)$; $\bar{d}(\{x_i\},\{y_i\}) = \Sigma_{i=1}^{n} d_i(x_i,y_i)$.

5. Let X be a non-empty set and f a real function defined on X. Show that f is bounded in the sense of the definition given in the last paragraph of the text $\Leftrightarrow$ there exists a real number K such that $|f(x)| \leq K$ for every $x \in X \Leftrightarrow \sup |f(x)| < +\infty$. Consider the set of all bounded real functions defined on X, and define the norm of a function f in this set by

$$\|f\| = \sup |f(x)|.$$

It is obvious that $\|f\|$ is a non-negative real number such that $\|f\| = 0 \Leftrightarrow f = 0$, and that $\|-f\| = \|f\|$. Prove in detail that $\|f + g\| \leq \|f\| + \|g\|$.

6. Let I be a subset of the real line. Show that I is an interval $\Leftrightarrow$ it is non-empty and contains each point between any two of its points (in the sense that if x and z are in I and $x \leq y \leq z$, then y is in I). If $\{I_i\}$ is a non-empty class of intervals on the real line such that $\cap_i I_i$ is non-empty, show that $\cup_i I_i$ is an interval.
7. Let X be a metric space with metric d. If x is a point of X and A a subset of X, show the following: if A is non-empty, $d(x,A)$ is a non-negative real number; and $d(x,A) = +\infty \Leftrightarrow A$ is empty.
8. Let X be a metric space with metric d and A a subset of X. Show the following: if A is non-empty, $d(A)$ is a non-negative extended real number; $d(A) = -\infty \Leftrightarrow A$ is empty; and if A is bounded, it is non-empty.

10. OPEN SETS

Let X be a metric space with metric d. If x_0 is a point of X and r is a positive real number, the *open sphere* $S_r(x_0)$ with *center* x_0 and *radius* r is the subset of X defined by

$$S_r(x_0) = \{x : d(x,x_0) < r\}.$$

An open sphere is always non-empty, for it contains its center. In Example 9-1, an open sphere with radius 1 contains only its center. $S_r(x_0)$ is often called the open sphere with radius r *centered on* x_0; intuitively, it consists of all points in X which are "close" to x_0, with the degree of closeness given by r.

A few concrete examples are in order. It should be easy to visualize the open sphere $S_r(x_0)$ on the real line: it is the bounded open interval $(x_0 - r, x_0 + r)$ with mid-point x_0 and total length $2r$. Conversely, it is clear that any bounded open interval on the real line is an open sphere,

so the open spheres on the real line are precisely the bounded open intervals. The open sphere $S_r(z_0)$ in the complex plane (see Fig. 17) is the inside of the circle with center z_0 and radius r. Figure 18 illustrates an open sphere in the space $\mathcal{C}[0,1]$: $S_r(f_0)$ consists of all functions f in $\mathcal{C}[0,1]$ whose graphs lie within the shaded band of vertical width $2r$ centered on the graph of f_0.

A subset G of the metric space X is called an *open set* if, given any point x in G, there exists a positive real number r such that $S_r(x) \subseteq G$,

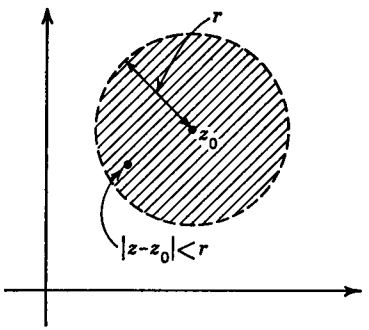


Fig. 17. An open sphere in the complex plane.

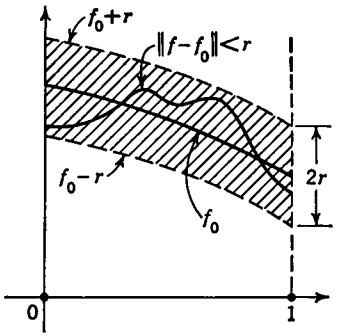


Fig. 18. An open sphere in $\mathcal{C}$ [0,1].

that is, if each point of G is the center of some open sphere contained in G. Loosely speaking, a set is open if each of its points is "inside" the set, in the sense made precise by the definition. On the real line, a set consisting of a single point is not open, for each bounded open interval centered on the point contains points not in the set. Similarly, the subset [0,1) of the real line is not open, because the point 0 in [0,1) has the property that each bounded open interval centered on it (no matter how small it may be) contains points not in [0,1), e.g., negative points. If we omit the offending point 0, the resulting bounded open interval (0,1) is an open set (this is very easy to prove and is a special case of Theorem B below). Further, it is quite clear that *any* open interval—bounded or not—is an open set, and also that the open intervals are the only intervals which are open sets.

Theorem A. *In any metric space X, the empty set $\emptyset$ and the full space X are open sets.*

PROOF. To show that $\emptyset$ is open, we must show that each point in $\emptyset$ is the center of an open sphere contained in $\emptyset$; but since there are no points in $\emptyset$, this requirement is automatically satisfied. X is clearly open, since every open sphere centered on each of its points is contained in X.

We have seen that [0,1) is not open as a subset of the real line. However, if we consider [0,1) as a metric space X in its own right, as a subspace of the real line, then [0,1) is open as a subset of X, since from this point of view it is the full space. This apparent paradox disappears when we realize that points outside of a given metric space have no relevance to any discussion taking place within the context of that space. A set is open or not open only with respect to a specific metric space containing it, never on its own.

Our next theorem justifies the adjective in the expression "open sphere."

Theorem B. *In any metric space X, each open sphere is an open set.*

PROOF. Let $S_r(x_0)$ be an open sphere in X, and let x be a point in $S_r(x_0)$. We must produce an open sphere centered on x and contained in $S_r(x_0)$. Since $d(x,x_0) < r$, $r_1 = r - d(x,x_0)$ is a positive real number. We show that $S_{r_1}(x) \subseteq S_r(x_0)$. If y is a point in $S_{r_1}(x)$, so that $d(y,x) < r_1$, then $d(y,x_0) \leq d(y,x) + d(x,x_0) < r_1 + d(x,x_0) = [r - d(x,x_0)] + d(x,x_0) = r$ shows that y is in $S_r(x_0)$.

The following characterization of open sets in terms of open spheres is a useful tool.

Theorem C. *Let X be a metric space. A subset G of X is open $\Leftrightarrow$ it is a union of open spheres.*

PROOF. We assume first that G is open, and we show that it is a union of open spheres. If G is empty, it is the union of the empty class of open spheres. If G is non-empty, then since it is open, each of its points is the center of an open sphere contained in it, and it is the union of all the open spheres contained in it.

We now assume that G is the union of a class **S** of open spheres. We must show that G is open. If **S** is empty, then G is also empty, and by Theorem A, G is open. Suppose that **S** is non-empty. G is also non-empty. Let x be a point in G. Since G is the union of the open spheres in **S**, x belongs to an open sphere $S_r(x_0)$ in **S**. By Theorem B, x is the center of an open sphere $S_{r_1}(x) \subseteq S_r(x_0)$. Since $S_r(x_0) \subseteq G$, $S_{r_1}(x) \subseteq G$ and we have an open sphere centered on x and contained in G. G is therefore open.

The fundamental properties of the open sets in a metric space are those stated in

Theorem D. *Let X be a metric space. Then* (1) *any union of open sets in X is open; and* (2) *any finite intersection of open sets in X is open.*

PROOF. To prove (1), let $\{G_i\}$ be an arbitrary class of open sets in X. We must show that $G = \cup_i G_i$ is open. If $\{G_i\}$ is empty, then G is

empty, and by Theorem A, G is open. Suppose that $\{G_i\}$ is non-empty. By Theorem C, each G_i (being an open set) is a union of open spheres; G is the union of all the open spheres which arise in this way; and by another application of Theorem C, G is open.

To prove (2), let $\{G_i\}$ be a finite class of open sets in X. We must show that $G = \bigcap_i G_i$ is open. If $\{G_i\}$ is empty, then $G = X$; and by Theorem A, G is open. Suppose that $\{G_i\}$ is non-empty and that $\{G_i\} = \{G_1, G_2, \ldots, G_n\}$ for some positive integer n. If G happens to be empty, then it is open by Theorem A, so we may assume that G is non-empty. Let x be a point in G. Since x is in each G_i, and each G_i is open, for each i there is a positive real number r_i such that $S_{r_i}(x) \subseteq G_i$. Let r be the smallest number in the set $\{r_1, r_2, \ldots, r_n\}$. This number r is a positive real number such that $S_r(x) \subseteq S_{r_i}(x)$ for each i, so $S_r(x) \subseteq G_i$ for each i, and therefore $S_r(x) \subseteq G$. Since $S_r(x)$ is an open sphere centered on x and contained in G, G is open.

The above theorem says that the class of all open sets in a metric space is closed under the formation of arbitrary unions and finite intersections. The reader should clearly understand that Theorem A is an immediate consequence of this statement, since the empty set is the union of the empty class of open sets and the full space is its intersection. The limitation to *finite* intersections in this theorem is essential. To see this, it suffices to consider the following sequence of open intervals on the real line:

$$(-1,1),\ (-\tfrac{1}{2},\tfrac{1}{2}),\ (-\tfrac{1}{3},\tfrac{1}{3}),\ \ldots$$

The intersection of these open sets is the set $\{0\}$ consisting of the single point 0, and this set is not open.

In an arbitrary metric space, the structure of the open sets can be very complicated indeed. Theorem C contains the best information available in the general case: each open set is a union of open spheres. In the case of the real line, however, a description can be given of the open sets which is fairly explicit and reasonably satisfying to the intuition.

Theorem E. *Every non-empty open set on the real line is the union of a countable disjoint class of open intervals.*

PROOF. Let G be a non-empty open subset of the real line. Let x be a point of G. Since G is open, x is the center of a bounded open interval contained in G. Define I_x to be the union of all the open intervals which contain x and are contained in G. The following three facts are easily proved: I_x is an open interval (by Theorem D and Problem 9-6) which contains x and is contained in G; I_x contains every open interval which contains x and is contained in G; and if y is another point in I_x,

then $I_x = I_y$. We next observe that if x and y are any two distinct points of G, then I_x and I_y are either disjoint or identical; for if they have a common point z, then $I_x = I_z$ and $I_y = I_z$, so $I_x = I_y$. Consider the class $\mathfrak{I}$ of all distinct sets of the form I_x for points x in G. This is a disjoint class of open intervals, and G is obviously its union. It remains to be proved that $\mathfrak{I}$ is countable. Let G_r be the set of rational points in G. G_r is clearly non-empty. We define a mapping f of G_r onto $\mathfrak{I}$ as follows: for each r in G_r, let $f(r)$ be that unique interval in $\mathfrak{I}$ which contains r. G_r is countable by Problem 6-7, and the fact that $\mathfrak{I}$ is countable follows from Problem 6-8.

A firm grasp of the ideas involved in the theory of metric spaces depends on one's capacity to "see" these spaces with the mind's eye. The complex plane is perhaps the best metric space to use as a model from which to absorb this necessary intuitive understanding. When we consider an unspecified set A of complex numbers, we usually imagine it as a region bounded by a curve, as in Fig. 19. We think of the point z_1, which is completely surrounded by points of A, as being "inside" the set A, or in its "interior," while z_2 is on the "boundary" of A. More precisely, z_1 is the center of some open sphere contained in A, and each open sphere centered on z_2 intersects both A and its complement A'. We formulate these ideas for a general metric space in the next paragraph and at the end of the next section.

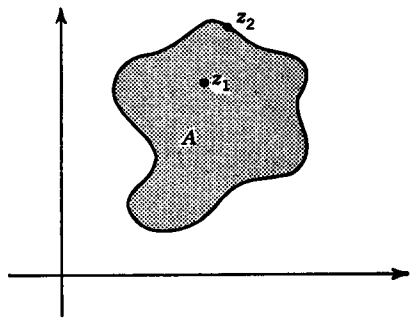


Fig. 19. A set A of complex numbers with interior point z_1 and boundary point z_2.

Let X be an arbitrary metric space, and let A be a subset of X. A point in A is called an *interior point* of A if it is the center of some open sphere contained in A; and the *interior* of A, denoted by $\mathrm{Int}(A)$, is the set of all its interior points. Symbolically,

$$\mathrm{Int}(A) = \{x : x \in A \text{ and } S_r(x) \subseteq A \text{ for some } r\}.$$

The basic properties of interiors are the following:

(1) $\mathrm{Int}(A)$ is an open subset of A which contains every open subset of A (this is often expressed by saying that the interior of A is the largest open subset of A);

(2) A is open $\Leftrightarrow A = \mathrm{Int}(A)$;

(3) $\mathrm{Int}(A)$ equals the union of all open subsets of A.

The proofs of these facts are quite easy, and we ask the reader to fill in the details as an exercise (see Problem 8).

Problems

1. Let X be a metric space, and show that any two distinct points of X can be separated by open spheres in the following sense: if x and y are distinct points in X, then there exists a disjoint pair of open spheres each of which is centered on one of the points.
2. Let X be a metric space. If $\{x\}$ is a subset of X consisting of a single point, show that its complement $\{x\}'$ is open. More generally, show that A' is open if A is any finite subset of X.
3. Let X be a metric space and $S_r(x)$ the open sphere in X with center x and radius r. Let A be a subset of X with diameter less than r which intersects $S_r(x)$. Prove that $A \subseteq S_{2r}(x)$.
4. Let X be a metric space. Show that every subset of X is open $\Leftrightarrow$ each subset of X which consists of a single point is open.
5. Let X be a metric space with metric d, and let d_1 be the metric defined in Problem 9-1. Show that the two metric spaces (X,d) and (X,d_1) have precisely the same open sets. (*Hint:* show that they have the same open spheres with one exception. What is this exception?)
6. If $X = X_1 \times X_2 \times \cdots \times X_n$ is the product in Problem 9-4, and if d and $\bar{d}$ are the metrics on X defined in that problem, show that the two metric spaces (X,d) and $(X,\bar{d})$ have precisely the same open sets. Observe that in this case the spaces do not have the same open spheres.
7. Let Y be a subspace of a metric space X, and let A be a subset of the metric space Y. Show that A is open as a subset of $Y \Leftrightarrow$ it is the intersection with Y of a set which is open in X.
8. Prove the statements made in the text about interiors.
9. Describe the interior of each of the following subsets of the real line: the set of all integers; the set of all rationals; the set of all irrationals; $(0,1)$; $[0,1]$; $[0,1) \cup \{1,2\}$. Do the same for each of the following subsets of the complex plane: $\{z:|z| < 1\}$; $\{z:|z| \leq 1\}$; $\{z:I(z) = 0\}$; $\{z:R(z)$ is rational$\}$.
10. Let A and B be two subsets of a metric space X, and prove the following:
    (*a*) $\text{Int}(A) \cup \text{Int}(B) \subseteq \text{Int}(A \cup B)$;
    (*b*) $\text{Int}(A) \cap \text{Int}(B) = \text{Int}(A \cap B)$.
    Give an example of two subsets A and B of the real line such that $\text{Int}(A) \cup \text{Int}(B) \neq \text{Int}(A \cup B)$.

11. CLOSED SETS

Let X be a metric space with metric d. If A is a subset of X, a point x in X is called a *limit point* of A if each open sphere centered on x contains at least one point of A different from x. The essential idea here is that the points of A different from x get "arbitrarily close" to x, or "pile up" at x.

The subset $\{1, \frac{1}{2}, \frac{1}{3}, \ldots\}$ of the real line has 0 as a limit point; in fact, 0 is its only limit point. The closed-open interval $[0,1)$ has 0 as a limit point which is in the set and 1 as a limit point which is not in the set; further, every real number x such that $0 < x < 1$ is also a limit point of this set. The set of all integral points on the real line has no limit points at all, whereas every real number is a limit point of the set of all rationals. In Example 9-1, every open sphere of radius less than 1 consists only of its center, so no subset of this space has any limit points.

A subset F of the metric space X is called a *closed set* if it contains each of its limit points. In rough terms, a set is closed if its points do not get arbitrarily close to any point outside of it. Among the subsets of the real line mentioned in the preceding paragraph, only the set of integral points is closed. In Example 9-1, every subset is closed.

Theorem A. *In any metric space X, the empty set $\emptyset$ and the full space X are closed sets.*

PROOF. The empty set has no limit points, so it contains them all and is therefore closed. Since the full space X contains all points, it automatically contains its own limit points and thus is closed.

The following theorem characterizes closed sets in terms of open sets. We already know a good deal about open sets, so this characterization provides us with a useful tool for establishing properties of closed sets.

Theorem B. *Let X be a metric space. A subset F of X is closed $\Leftrightarrow$ its complement F' is open.*

PROOF. Assume first that F is closed. We show that F' is open. If F' is empty, it is open by Theorem 10-A, so we may suppose that F' is non-empty. Let x be a point in F'. Since F is closed and x is not in F, x is not a limit point of F. Since x is not in F and is not a limit point of F, there exists an open sphere $S_r(x)$ which is disjoint from F. $S_r(x)$ is an open sphere centered on x and contained in F', and since x was taken to be any point of F', F' is open.

We now assume that F' is open and show that F is closed. The only way F can fail to be closed is to have a limit point in F'. This cannot

happen, for since F' is open, each of its points is the center of an open sphere disjoint from F, and no such point can be a limit point of F.

If x_0 is a point in our metric space X, and r is a non-negative real number, the *closed sphere* $S_r[x_0]$ with *center* x_0 and *radius* r is the subset of X defined by

$$S_r[x_0] = \{x : d(x,x_0) \leq r\}.$$

$S_r[x_0]$ contains its center, and when $r = 0$ it contains only its center. The closed spheres on the real line are precisely the closed intervals. In this connection, we observe that though open spheres on the real line are open intervals, there are open intervals which are not open spheres, e.g., $(-\infty, +\infty)$.

The following theorem justifies the adjective in the phrase "closed sphere."

Theorem C. *In any metric space X, each closed sphere is a closed set.*

PROOF. Let $S_r[x_0]$ be a closed sphere in X. By Theorem B, it suffices to show that its complement $S_r[x_0]'$ is open. $S_r[x_0]'$ is open if it is empty, so we may assume that it is non-empty. Let x be a point in $S_r[x_0]'$. Since $d(x,x_0) > r$, $r_1 = d(x,x_0) - r$ is a positive real number. We take r_1 as the radius of an open sphere $S_{r_1}(x)$ centered on x, and we show that $S_r[x_0]'$ is open by showing that $S_{r_1}(x) \subseteq S_r[x_0]'$. Let y be a point in $S_{r_1}(x)$, so that $d(y,x) < r_1$. On the basis of this and the fact that $d(x_0,x) \leq d(x_0,y) + d(y,x)$, we see that

$$d(y,x_0) \geq d(x,x_0) - d(y,x) > d(x,x_0) - r_1 = d(x,x_0) - [d(x,x_0) - r] = r,$$

so that y is in $S_r[x_0]'$.

The main general facts about closed sets are those given in our next theorem.

Theorem D. *Let X be a metric space. Then* (1) *any intersection of closed sets in X is closed; and* (2) *any finite union of closed sets in X is closed.*

PROOF. By virtue of Eqs. 2-(2) and Theorem B above, this theorem is an immediate consequence of Theorem 10-D. We prove (1) as follows. If $\{F_i\}$ is an arbitrary class of closed subsets of X and $F = \bigcap_i F_i$, then by Theorem B, F is closed if F' is open; but $F' = \bigcup_i F_i'$ is open by Theorem 10-D, since by Theorem B each F_i' is open. The second statement is proved similarly.

In Theorem E of the previous section, we gave an explicit characterization of the open sets on the real line. We now consider the structure of its closed sets. Among the simplest closed sets on the real line are the closed intervals (which are the closed spheres) and finite unions

of closed intervals. Finite sets are included among these, since a set consisting of a single point is a closed interval with equal end-points. What is the character of the most general closed set on the real line? Since closed sets are the complements of open sets, Theorem 10-E gives a complete answer to this question: the most general proper closed subset of the real line is obtained by removing a countable disjoint class of open intervals. This process sounds innocent enough, but in fact it leads to some rather curious and complicated examples. One of these examples is of particular importance. It was studied by Cantor and is usually called the *Cantor set.*

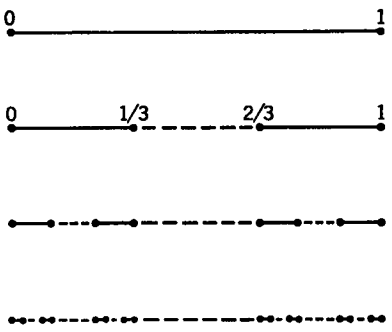


Fig. 20. The Cantor set.

To construct the Cantor set, we proceed as follows (see Fig. 20). First, denote the closed unit interval [0,1] by F_1. Next, delete from F_1 the open interval $(\frac{1}{3},\frac{2}{3})$ which is its middle third, and denote the remaining closed set by F_2. Clearly,

$$F_2 = [0,\tfrac{1}{3}] \cup [\tfrac{2}{3},1].$$

Next, delete from F_2 the open intervals $(\frac{1}{9},\frac{2}{9})$ and $(\frac{7}{9},\frac{8}{9})$, which are the middle thirds of its two pieces, and denote the remaining closed set by F_3. It is easy to see that

$$F_3 = [0,\tfrac{1}{9}] \cup [\tfrac{2}{9},\tfrac{1}{3}] \cup [\tfrac{2}{3},\tfrac{7}{9}] \cup [\tfrac{8}{9},1].$$

If we continue this process, at each stage deleting the open middle third of each closed interval remaining from the previous stage, we obtain a sequence of closed sets F_n, each of which contains all its successors. The Cantor set F is defined by

$$F = \bigcap_{n=1}^{\infty} F_n,$$

and it is closed by Theorem *D*. F consists of those points in the closed unit interval [0,1] which "ultimately remain" after the removal of all the open intervals $(\frac{1}{3},\frac{2}{3})$, $(\frac{1}{9},\frac{2}{9})$, $(\frac{7}{9},\frac{8}{9})$, ... What points do remain? F clearly contains the end-points of the closed intervals which make up each set F_n:

$$0,\ 1,\ \tfrac{1}{3},\ \tfrac{2}{3},\ \tfrac{1}{9},\ \tfrac{2}{9},\ \tfrac{7}{9},\ \tfrac{8}{9},\ \ldots$$

Does F contain any other points? We leave it to the reader to verify that $\frac{1}{4}$ is in F and is not an end-point. Actually, F contains a multitude of points other than the above end-points, for the set of these end-points

is clearly countable, while the cardinal number of F itself is c, the cardinal number of the continuum. To prove this, it suffices to exhibit a one-to-one mapping f of $[0,1)$ into F. We construct such a mapping as follows. Let x be a point in $[0,1)$, and let $x = .b_1b_2b_3 \ldots$ be its binary expansion (see Sec. 7). Each b_n is either 0 or 1. Let $t_n = 2b_n$, and regard $.t_1t_2t_3 \ldots$ as the ternary expansion of a real number $f(x)$ in $[0,1)$. The reader will easily convince himself that $f(x)$ is in the Cantor set F: since t_1 is 0 or 2, $f(x)$ is not in $[\frac{1}{3},\frac{2}{3})$; since t_2 is 0 or 2, $f(x)$ is not in $[\frac{1}{9},\frac{2}{9})$ or $[\frac{7}{9},\frac{8}{9})$; etc. Also, it is easy to see that the mapping $f\colon [0,1) \to F$ is one-to-one. According to this, F contains exactly as many points as the entire closed unit interval $[0,1]$. It is interesting to compare this conclusion with the fact that the sum of the lengths of all the open intervals removed is precisely 1, since

$$\tfrac{1}{3} + \tfrac{2}{9} + \tfrac{4}{27} + \cdots = 1.$$

It is also interesting to observe (by doing a little arithmetic) that F_{25} is the union of 16,777,216 disjoint closed intervals of the same length which are rather irregularly distributed along $[0,1]$. These facts may suffice to indicate that the Cantor set is a very intricate mathematical object and is just the sort of thing mathematicians delight in. We shall encounter this set again from time to time, for its properties illustrate several phenomena discussed in later sections.

We conclude this section by defining two additional concepts which are often useful.

Let X be an arbitrary metric space, and let A be a subset of X. The *closure* of A, denoted by $\bar{A}$, is the union of A and the set of all its limit points. Intuitively, $\bar{A}$ is A itself together with all other points in X which are arbitrarily close to A. As an example, if A is the open unit disc $\{z\colon |z| < 1\}$ in the complex plane, then $\bar{A}$ is the closed unit disc $\{z\colon |z| \leq 1\}$. The main facts about closures are the following:

(1) $\bar{A}$ is a closed superset of A which is contained in every closed superset of A (we express this by saying that $\bar{A}$ is the smallest closed superset of A);

(2) A is closed $\Leftrightarrow A = \bar{A}$;

(3) $\bar{A}$ equals the intersection of all closed supersets of A.

It is a routine exercise to prove these statements, and we leave this task to the reader (in Problem 6).

Our second concept relates to the discussion of Fig. 19 given at the end of the previous section. Again, let X be a metric space and A a subset of X. A point in X is called a *boundary point* of A if each open sphere centered on the point intersects both A and A', and the *boundary* of A is the set of all its boundary points. This concept possesses the following properties:

(1) the boundary of A equals $\bar{A} \cap \overline{A'}$;
(2) the boundary of A is a closed set;
(3) A is closed $\Leftrightarrow$ it contains its boundary.

We ask the reader to give the proofs in Problem 11.

Problems

1. Let X be a metric space, and extend Problem 10-1 by proving the following statements:
   (*a*) any point and disjoint closed set in X can be separated by open sets, in the sense that if x is a point and F a closed set which does not contain x, then there exists a disjoint pair of open sets G_1 and G_2 such that $x \in G_1$ and $F \subseteq G_2$;
   (*b*) any disjoint pair of closed sets in X can be separated by open sets, in the sense that if F_1 and F_2 are disjoint closed sets, then there exists a disjoint pair of open sets G_1 and G_2 such that $F_1 \subseteq G_1$ and $F_2 \subseteq G_2$.
2. Let X be a metric space, and let A be a subset of X. If x is a limit point of A, show that each open sphere centered on x contains an infinite number of distinct points of A. Use this result to show that a finite subset of X is closed.
3. Show that a subset of a metric space is bounded $\Leftrightarrow$ it is non-empty and is contained in some closed sphere.
4. Give an example of an infinite class of closed sets whose union is not closed. Give an example of a set which (*a*) is both open and closed; (*b*) is neither open nor closed; (*c*) contains a point which is not a limit point of the set; and (*d*) contains no point which is not a limit point of the set.
5. Describe the interior of the Cantor set.
6. Prove the statements made in the text about closures.
7. Let X be a metric space and A a subset of X. Prove the following facts:
   (*a*) $\bar{A}' = \text{Int}(A')$;
   (*b*) $\bar{A} = \{x : d(x,A) = 0\}$.
8. Describe the closure of each of the following subsets of the real line: the integers; the rationals; the Cantor set; $(0,+\infty)$; $(-1,0) \cup (0,1)$. Do the same for each of the following subsets of the complex plane: $\{z : |z| \text{ is rational}\}$; $\{z : 1/R(z) \text{ is an integer}\}$; $\{z : |z| < 1 \text{ and } I(z) < 0\}$.
9. Let X be a metric space, let x be a point of X, and let r be a positive real number. One is inclined to believe that the closure of $S_r(x)$ must equal $S_r[x]$. Give an example to show that this is not necessarily true. (*Hint:* see Example 9-1.)

10. Let X be a metric space, and let G be an open set in X. Prove that G is disjoint from a set $A \Leftrightarrow G$ is disjoint from $\bar{A}$.
11. Prove the facts about boundaries stated in the text.
12. Describe the boundary of each of the following subsets of the real line: the integers; the rationals; $[0,1]$; $(0,1)$. Do the same for each of the following subsets of the complex plane: $\{z:|z| < 1\}$; $\{z:|z| \leq 1\}$; $\{z:I(z) > 0\}$.
13. Let X be a metric space and A a subset of X. A is said to be *dense* (or *everywhere dense*) if $\bar{A} = X$. Prove that A is dense $\Leftrightarrow$ the only closed superset of A is $X \Leftrightarrow$ the only open set disjoint from A is $\emptyset \Leftrightarrow A$ intersects every non-empty open set $\Leftrightarrow A$ intersects every open sphere.

12. CONVERGENCE, COMPLETENESS, AND BAIRE'S THEOREM

As we emphasized in the introduction to this chapter, one of our main aims in considering metric spaces is to study convergent sequences in a context more general than that of classical analysis. The fruits of this study are many, and among them is the added insight gained into ordinary convergence as it is used in analysis.

Let X be a metric space with metric d, and let

$$\{x_n\} = \{x_1, x_2, \ldots, x_n, \ldots\}$$

be a sequence of points in X. We say that $\{x_n\}$ is *convergent* if there exists a point x in X such that either

(1) for each $\epsilon > 0$, there exists a positive integer n_0 such that $n \geq n_0 \Rightarrow d(x_n,x) < \epsilon$; or equivalently,

(2) for each open sphere $S_\epsilon(x)$ centered on x, there exists a positive integer n_0 such that x_n is in $S_\epsilon(x)$ for all $n \geq n_0$.

The reader should observe that the first condition is a direct generalization of convergence for sequences of numbers as defined in the introduction, and that the second can be thought of as saying that each open sphere centered on x contains all points of the sequence from some place on. If we rely on our knowledge of what is meant by a convergent sequence of real numbers, the statement that $\{x_n\}$ is convergent can equally well be defined as follows: there exists a point x in X such that $d(x_n,x) \to 0$. We usually symbolize this by writing

$$x_n \to x,$$

and we express it verbally by saying that x_n *approaches* x, or that x_n *converges to* x. It is easily seen from condition (2) and Problem 10-1 that the point x in this discussion is unique, that is, that $x_n \to y$ with $y \neq x$